Chang Liu

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Education

- **University of Ottawa, Ottawa, ON, Canada** *Sept.2022 - August.2024(Expected)*
  Master of Computer Science and Concentration Applied Artificial Intelligence CGPA: 9.6/10
- **Carleton University, Ottawa, ON, Canada** *Sept.2017 - August.2022*
  Bachelor of Computer Science Honours CGPA: 10.18/12

Experience

- **IBM** *Feb.2023 - Present*
  CAS Project Student Ottawa, ON
  – Collaborated with DBMS experts to improve my work on using AI to improve the query optimizer (especially in join order selection).
  – Delivered ideas and results to DBMS experts and improve the performance of my model based on their feedback
- **Ford Motor Company** *Sept.2020 - Dec.2021*
  Test Automation - Co-op Ottawa, ON
  – Conduct automated software testing using python for the electronic control units (ECUs) in Ford vehicles through Jenkins.
  – Participate in continuous quality improvement initiatives, including root cause analysis, defects triage, fix validation.
  – Acquired working proficiencies in issue tracking and project management with Jira
- **Carleton University / University of Ottawa** *Sept.2020 - Apr.2024*
  Teaching Assistant Ottawa, ON
  – Conduct seminars, discussion groups and laboratory sessions to supplement lectures.
  – Grade examinations and assignment.

Projects

- **Join Order Selection Using Deep Reinforcement Learning** *May.2023 - present*
  Master Thesis, University of Ottawa
  – **Objective:** Developed an advanced query optimizer using deep reinforcement learning for relational databases to enhance query execution efficiency by determining the optimal join order.
  – **Technologies Used:** Python, PyTorch, PostgreSQL, Tree-LSTMs, GNNs, DRL
  – **Key Contributions:**
    * Designed and implemented a novel framework combining Tree-LSTMs and GCNs to effectively encode hierarchical and relational data structures of SQL queries.
    * Employed Dueling-DQN to learn optimal join orders, optimizing for minimal execution costs.
    * Conducted extensive benchmarking against traditional query optimizers using standard datasets, demonstrating superior performance in query execution time and cost.
    * Achieved robust generalization, with the model effectively optimizing previously unseen queries.

Technical Skills and Interests

Technical Skills: Python, Relational Databases (PostgreSQL), Pytorch, Numpy, Data Visualization, SciKit Learn, Jupyter Notebook, C++, Java, Android Studio
Soft Skills: Critical thinking, Prioritization, Self-disciplined, Self-movivated
Field of Interest: Deep Learning, Machine Learning, Reinforcement Learning

Positions of Responsibility

- **Freshman Support Group Manager** Carleton University *Sept.2020 - Dec.2020*
  Came up with practical themes for weekly group meetings, such as time and stress management, information gathering, maintaining academic integrity, achieving study-life balance, and staying positive.
- **Co-op Trainer** Ford Motor Company *Sept.2022 - Dec.2022*
  Delivered induction training to co-op interns, including daily work procedures and providing guidance and support to ensure successful integration into the team.

Achievements

- **Deans' Honour List** Carleton University *2020*
- **Murdoch Maxwell MacOdrum Scholarship** Carleton University *2020*